

# Glossary

**ACID** A corrosive substance that will react with a base (see base) to form a salt and water.

**ALCHEMY** Practices in medieval times that tried to turn common metals, such as lead, into gold.

**ALKALI** A soluble base (see base).

**ALLOY** A mixture of two or more metals.

**ATMOSPHERIC PRESSURE** The pressure created by the gases surrounding Earth.

**ATOM** The smallest unit of an element.

**ATOMIC MASS** The mass of an atom measured in atomic mass units (AMU).

**BASE** A corrosive substance that will react with an acid to form a salt and water.

**BLOCK** Larger sets of elements on the periodic table, which may have similar properties.

**BOND** The attraction between atoms or ions that holds them together in an element or a compound.

**CATALYST** A substance that can be added to a chemical reaction to speed it up.

**COMBUSTION** A chemical reaction in which a fuel burns in the presence of oxygen.

**COMPOUND** A substance made up of elements bonded together.

**CONDUCTIVITY** The ability of a substance to let electricity, sound, or heat pass through it.

**CORROSION** The process of a metal breaking down by chemically reacting with substances in the environment. An example of this is the rusting of iron through reactions with oxygen and water.

**CYCLOTRON** A type of particle accelerator. Particle accelerators are machines used to smash atoms into one another at incredibly high speeds.

**ELECTROLYSIS** A technique that uses electricity to break salt compounds down into simpler substances.

**ELECTRON** A negatively charged particle that orbits the nucleus of an atom.

**ELEMENT** A substance that cannot be broken down into anything simpler.

**EMISSION** The process of giving off light, heat, radiation, or particles.

**FISSION** The process of splitting the nucleus of an atom into smaller fragments.

**FOSSIL FUEL** A substance formed in Earth's crust from the remains of ancient organisms. Humans use this as fuel.

**FUEL CELL** A device that converts chemical energy into electricity.

**FUSION** Where two or more atomic nuclei combine to form different atomic nuclei.

**GALVANIZATION** The process of electrically coating steel with zinc to protect it from corrosion.

**GENETICS** The study of inherited traits between different organisms.

**GROUP** A vertical column of elements on the periodic table, with elements that often have similar properties.

**HALF-LIFE** The time taken for half of the nuclei in a radioactive sample to undergo decay (see nuclear decay).

**INERT** A way of describing a substance that does not easily undergo a chemical reaction.

**INSULATOR** A substance that does not let electricity, sound, or heat flow through it easily.

**ION** A charged particle produced when an atom either loses or gains electrons.

**ISOTOPE** A form of an element with atoms that have the same number of protons but a different number of neutrons.

**MALLEABILITY** A metal's ability to be reshaped.

**MELTING POINT** The temperature at which a solid gets hot enough to turn into a liquid.

**METAL** An element that is usually found as a shiny solid and is a good conductor of electricity and heat.



Crystals of the element bismuth

**METALLOID** An element that displays properties of both metals and non-metals.

**MINERAL** A naturally occurring form of a chemical compound, usually found in Earth's crust.

**MOLECULE** A group of atoms that are chemically bonded to one another.

**NEUTRON** A neutral subatomic particle found in the nucleus of an atom.

**NUCLEAR BOMB** An explosive device that releases large amounts of energy and can cause mass devastation.

**NUCLEAR DECAY** A natural process in which the nuclei of radioactive atoms break down or rearrange themselves to form new nuclei.

**NUCLEAR REACTOR** A device used to carry out a controlled nuclear reaction.



Saxophone made of bronze, a copper-tin alloy



Rust on motorcycle, an example of corrosion